1 Integrals of $\sin(x)$ and $\cos(x)$

1. Use the derivative formulas $\frac{d}{dx}\sin(x) = \cos(x)$ and $\frac{d}{dx}\cos(x) = -\sin(x)$ to find $\int \sin(x)dx$ and $\int \cos(x)dx$. Check your answers by differentiating.

   (a) $\int \sin(x)dx =$

   (b) $\int \cos(x)dx =$

2 Integrals of $\tan(x)$ and $\cot(x)$

2. Write $\tan(x) = \frac{\sin(x)}{\cos(x)}$ and $\cot(x) = \frac{\cos(x)}{\sin(x)}$ and use substitution to find $\int \tan(x)dx$ and $\int \cot(x)dx$. Check your answers by differentiating.

   (a) $\int \tan(x)dx =$

   (b) $\int \cot(x)dx =$

3 Integrals of $\sec(x)$ and $\csc(x)$

3. Evaluate $\int \sec(x)dx$. Check your answer by differentiating.
Hint: $\int \sec(x)dx = \int \sec(x) \cdot \frac{\sec(x)+\tan(x)}{\sec(x)+\tan(x)}dx$.

4. Evaluate $\int \csc(x)dx$. Check your answer by differentiating. *Hint:* Use a similar trick to the one above.

4 Exercises

5. Evaluate each integral.

(a) $\int \frac{\sec^2(1/x)}{x^2} dx$

(b) $\int \frac{\sin(\sqrt{x})}{\sqrt{x}} dx$

(c) $\int \sqrt{x}\cot(1+x^{3/2}) dx$

(d) $\int e^{\tan(x)} \sec^2(x) dx$

(e) $\int \sqrt{\cot(x)} \csc^2(x) dx$

(f) $\int \frac{\sec(\ln(x))}{x} dx$